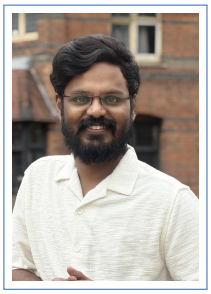
Akarsh K Nair

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Work Experience

Nov 2025 **Nottingham Trent University**, *Research Fellow*, Nottingham, United Kingdom.
HORIZON Europe Project Phase IV AI: Lung cancer detection using CPRD EHR Data
Under the guidance of Prof. David Brown, Department of CS ***[Ongoing Tenure]***

8/25 – 1/26 **Loughborough University** , *Visiting Researcher*, Loughborough, United Kingdom .
Under the Guidance of : Dr. Peng Liu , Dept of Mathetical Sciences

2024 – 2025 **Indian Institute of Information Technology Kottayam**, *Adhoc Faculty(fixed-term teaching contract)*, Kottayam, Kerala , India.

2023 – 2024 **University of Adger**, *Visiting Researcher*, WISENET Group, Dept. of ICT, UiA, Adger , Norway.
Project: Indo-Norwegian Collaboration in Autonomous Cyber-Physical Systems

2022– 2023 **iHub - HCI Foundation**, *Doctoral Fellow*, Indian Institute of Technology Mandi, HP, India.
Project: Exploring the applicability of Federated Learning for HCI applications

2022– 2023 **DADB Germany**, *Academic Assistant*, Remote Position.
Project: 5G Training & Labs

2019 – 2020 **TE College**, *Assistant Professor*, Dept. of Computer Science, Palakkad, Kerala, India (pre-doctoral appointment).

Education

2020 – 2024 **Indian Institute of Information Technology Kottayam**, Kottayam, Kerala, India
PhD in Computer Science, Advisor: *Dr. Jayakrushna Sahoo*
Thesis Title: Advancing Federated Learning Optimization: Privacy-Aware Efficient Multi-Tier Federated Learning Approaches for Mobile IoT Systems

2017 – 2019 **Christ College**, Thrissur, Affiliated to Calicut University, Kerala, India.

Master's Degree in Computer Science with First Class

2014 – 2017 **CAS**, Vadakkanchery, Affiliated to Calicut University, Kerala, India.
Bachelors's Degree in Electronics with First Class

Research Interest

Federated Learning, Edge Intelligence, Privacy Preservation and AI for Security

Selected Publications

TETCI 2025 **Akarsh K Nair**, Sinem Coleri, Linga Reddy Cenkeramaddi, Jayakrushna Sahoo, and Ebin Deni Raj. 2025. Incentivized Federated Learning: A Survey. In IEEE Transactions on Emerging Topics in Computational Intelligence, doi: 10.1109/TETCI.2025.3547609

IWCMC 2025 Arun Kumar Chavali, **Akarsh K Nair**, and Jayakrushna Sahoo. 2025. ComFLEX: A Communication efficient FL training framework for Edge computing environments. In IEEE The 21st International Wireless Communications & Mobile Computing Conference , doi:10.1109/IWCMC65282.2025.11059513

IRI 2025 **Akarsh K Nair**, Shanik Hubert , and Deepti Gupta. AndroIDS: Android-based Intrusion Detection System using Federated Learning. IEEE 26th International Conference on Information Reuse and Integration for Data Science

CPSS 2024 **Akarsh K Nair**, Jayakrushna Sahoo, Linga Reddy Cenkeramaddi, Gaurav Jaswal and Ebin Deni Raj. 2024. Mobile_FL: A streamlined FL framework for process optimisation via client clustering using rough c-means algorithm. In CPSS '24: Proceedings of the 10th ACM Cyber-Physical System Security Workshop (CPSS'24) , Singapore, 88 - 97.

CSI 2023 **Akarsh K Nair**, Jayakrushna Sahoo and Ebin Deni Raj. 2023. Privacy preserving Federated Learning framework for IoMT based big data analysis using edge computing. Volume 83, 103720, Elsevier Computer Standards & Interfaces, (April 2023).

CSI 2023 **Akarsh K Nair**, Ebin Deni Raj and Jayakrushna Sahoo. 2023. A robust analysis of adversarial attacks on federated learning environments. Volume 83, 103723, Elsevier Computer Standards & Interfaces, (April 2023).

Patent

2022 **Akarsh K Nair** and Jayakrushna Sahoo. AN EDGE COMPUTING-BASED INTELLIGENT PIPELINE FOR DATA OPTIMIZATION IN REMOTE CROWDSOURCING SYSTEMS. In INDIAN Patents (Application number: 202241056632).

2024 **Akarsh K Nair** , Jayakrushna Sahoo , Ebin Deni Raj , Afila A S and Sheema Madhusudhanan. PATHFINDER: A FEDERATED LEARNING BASED SECURE NAVIGATION SYSTEM FORVISUALLY IMPAIRED. In INDIAN Patents (Application number: 202441026960).

Achievements

Fellowship Selected for MSCA Postdoc Fellowship in Slovenia (MSCA SMASH)
Shortlisted for MSCA Postdoc Fellowship in Ireland (MSCA Rosetta)

Services

Reviewer IEEE TETCI, IEEE TIFS, IEEE TMC, IEEE TNSE
ACM Computing Surveys
Elsevier Future generation computer systems

Network Handles

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ORCID https://orcid.org/0000-0002-7734-0367
DBLP https://dblp.org/pid/347/6741.html

References

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   Professor, Dept. of Computer Science, NTU UK
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2. **Dr. Peng Liu**
   Senior Lecturer
   School of Mathematical Sciences, Loughborough University, UK
   Email : p.liu4@lboro.ac.uk
3. **Dr.Deepti Gupta**
   Assistant Professor, Dept. of CIS, Texas A&M University
   Email :d.gupta@tamuct.edu